

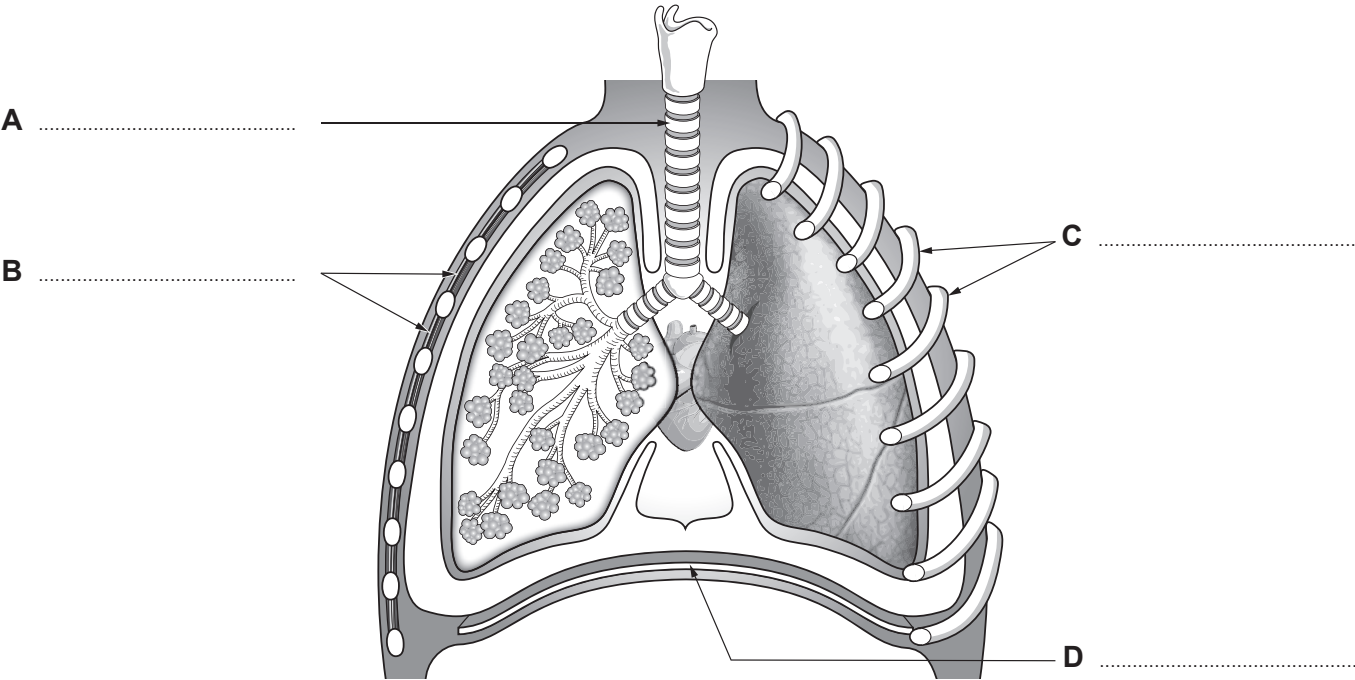
Examiner
only

5. Sports science students are learning about the respiratory system.

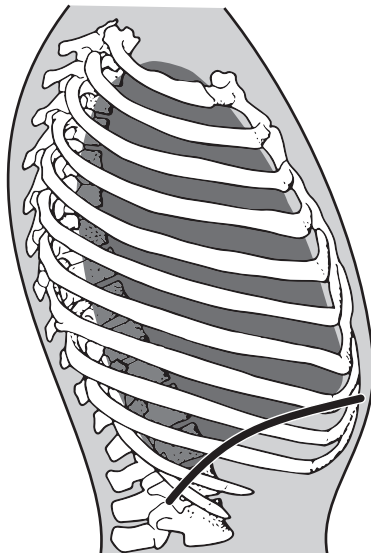
(a) (i) Label the diagram below using words from the box.

[4]

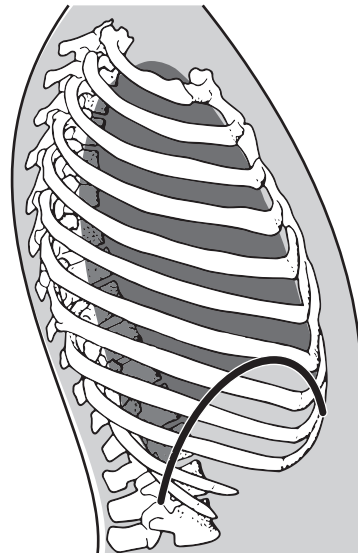
ribs intercostal muscles diaphragm lung trachea



[6 QER]



breathing in

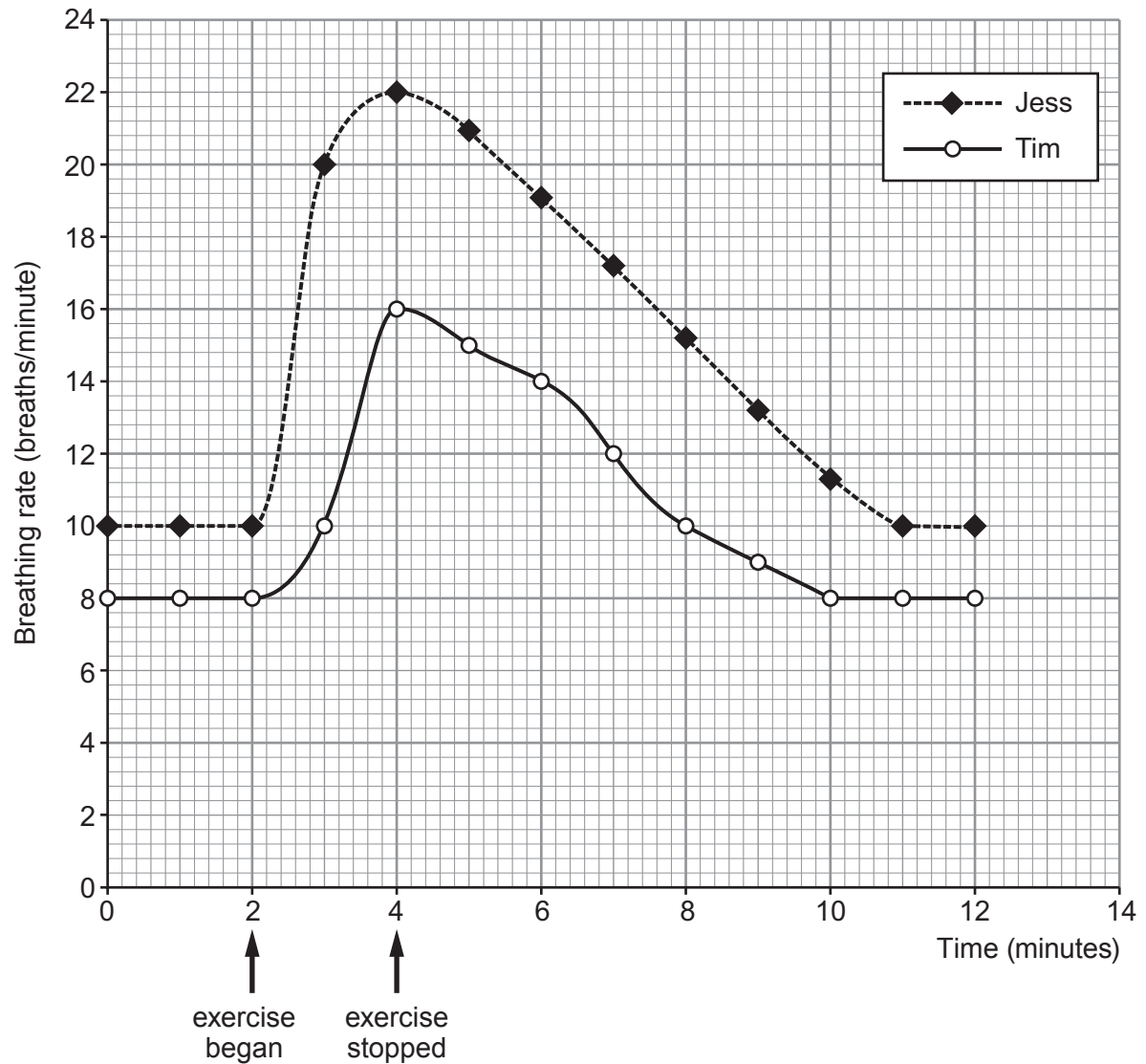


breathing out

- (b) Two students, Tim and Jess, studied the effect of exercise on breathing rate. They followed the method below:

1. record breathing rate before exercising
2. exercise for 2 minutes by running on a treadmill
3. measure the breathing rates during and after exercise up until 12 minutes

Their results are shown on the graph below.



Use the graph to complete the following sentences.

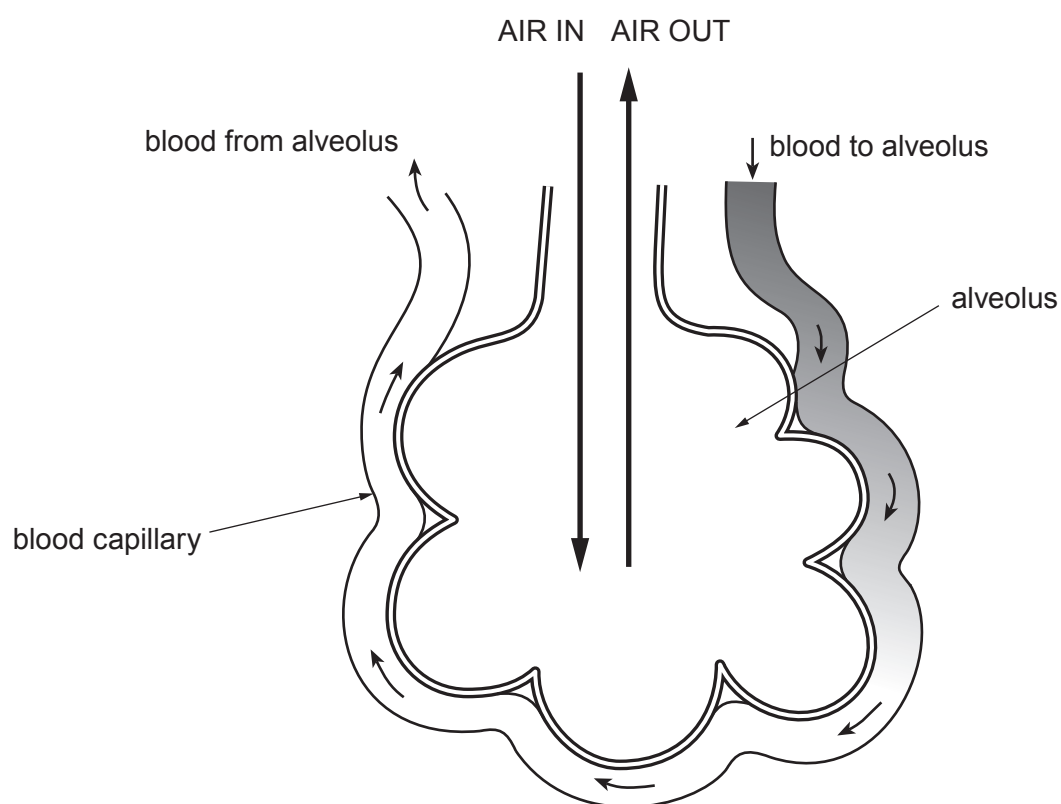
[3]

At 2 minutes, the difference between Tim and Jess' breathing rate is breaths per minute.

Between 8 and 10 minutes Tim's breathing rate drops than Jess' breathing rate.

Tim's breathing rate returns to normal at minutes.

- (c) The diagram below is of an alveolus within the lung where gases are exchanged.



- (i) State which gases are exchanged.

[2]

..... and

- (ii) Name the process by which these gases move between the alveolus and the capillary.

[1]

.....

- (iii) State **two** ways in which the alveoli are adapted for gas exchange.

[2]

1.

2.